

How GeoStats forecasts area prices

Method, safeguards and validation · Average sale price per m² across Dubai communities · Built 23 Sep 2026 from Dubai Land Department data to Aug 2025

119

COMMUNITIES FORECAST

24 mo

HORIZON, FROM SEP 2025

16.1%

TYPICAL ERROR (WAS 25% NAIVE)

85%

DIRECTION CALLED CORRECTLY

Why the method changed

The previous forecasts came from one gradient-boosted tree model (XGBoost) fitted to each area's raw monthly average price. Tree models cannot predict a value outside the range they were trained on, so they drifted back toward each area's 2010–2025 average: **82 of 115 areas were projected to fall** while the market was rising, and the lines read flat against visibly trending history. The replacement is built to extend a trend, to be tested before it is shown, and to stay silent where the data cannot support a forecast.

The method, in six steps

1. Clean the series	Cap the most extreme 2% of months at each end, then work in logarithms so every change is a percentage. One unusual villa sale can no longer steer an area.
2. Build one market line	For each month, the median price per m² across all reporting areas — roughly 96 areas a month, a far steadier signal than any single area.
3. Project that line	Damped trend smoothing (Holt): the market's direction continues but flattens gradually, so it neither runs away nor collapses to an average. Fitted settings α 0.5, β 0.2, damping 0.8.
4. Place each area on it	An area's price divided by the market line is its share — far more stable than its raw price. That share is smoothed and carried forward: area forecast = market forecast × area share .
5. Test before trusting	Rewind to each of the last 24 month-ends, refit using only the data available then, forecast 12 months, compare with what actually happened. If a plain trend line does better, that simpler line ships instead.
6. Publish the uncertainty	The shaded band is the 10th–90th percentile of this model's own past errors at each horizon, so it widens exactly as much as the model has actually been wrong.

Where it refuses to answer

A forecast is withheld rather than guessed. Of 151 communities with recorded sales, **22 have fewer than 12 recorded months** and **10 were wrong by more than 50% in testing** — their recorded prices mix property types too unevenly, one area alternating between AED 4,200 and AED 33,800 a month. Those 32 show recorded history only and the chart says no forecast is available. A further 75 communities have no residential sales at all in the bundled export.

What shipped, and how well it tests

Measure	Result
Market-and-share model used	90 areas — it beat the simple trend line in testing
Simple trend line used instead	31 areas
Typical error, model areas	16.1% (same areas with the simple line: 25.0%)
Typical error, trend-line areas	9.2%
Market index over the horizon	AED 14,981 → 15,382 per m² (+2.7%)
Areas rising / falling	91 rising, 29 falling — median +5.2%

A 16% typical error is why the band matters: the chart is read as a range, not as a single line. Each area's own error and method are printed on its Forecast card.

How it is checked

- Every one of the 121 published files was re-derived independently from the raw history and matched to better than 0.5%.
- A leakage test confirms that a fit using data up to any month is unchanged when later months are altered.
- All 2,856 forecast points drawn in the app were read back off the chart and matched the underlying data to 0.00%.
- Bands always contain the forecast, months are consecutive, and no forecast begins before the market's newest data.

How to run and read it

Refresh: after a new Dubai Land Department export, run `node build-price-index.js` then `node build-forecasts.js` — about five seconds, deterministic, no external service. The app does no modelling; it reads the published files.

Reading a chart: the solid line is recorded sales, the dashed line the forecast, the shaded band the likely range, and the badge states the typical error and which method that area used.

Other data points: each is registered with its own structure — counts on a log scale with seasonality, rates bounded to 0–100%, demographics on a long annual trend, composites recomputed from their parts. Only price per m² has a per-area history today; every other series must pass the same test before it is shown.

Source: Dubai Land Department export bundled with GeoStats — 151 areas, 14,139 recorded months, Jan 2010 to Aug 2025. An orientation for comparing areas, not a prediction of individual sale prices.